Chapter 19, Problem 9P

Problem

The *y* parameters of a network are:

$$Y = [y] = \begin{bmatrix} 0.5 & -0.2 \\ -0.2 & 0.4 \end{bmatrix} S$$

Determine the *z* parameters for the network.

Step-by-step solution

Step 1 of 5

Consider the following data:

The $\mathbf{y}$ parameters:

$$\begin{bmatrix} \mathbf{y}_{11} & \mathbf{y}_{12} \\ \mathbf{y}_{21} & \mathbf{y}_{22} \end{bmatrix} = \begin{bmatrix} 0.5 & -0.2 \\ -0.2 & 0.4 \end{bmatrix} \mathrm{S}$$

Step 2 of 5

Write the $\mathbf{z}$ parameters in terms of $\mathbf{y}$ parameter.

$$\begin{bmatrix} \mathbf{z}_{11} & \mathbf{z}_{12} \\ \mathbf{z}_{21} & \mathbf{z}_{22} \end{bmatrix} = \begin{bmatrix} \frac{\mathbf{y}_{22}}{\Delta_y} & -\frac{\mathbf{y}_{12}}{\Delta_y} \\ -\frac{\mathbf{y}_{21}}{\Delta_y} & \frac{\mathbf{y}_{11}}{\Delta_y} \end{bmatrix}$$

Calculate the value of Δ_y.

$$\begin{aligned} \Delta_y &= \mathbf{y}_{11}\mathbf{y}_{22} - \mathbf{y}_{12}\mathbf{y}_{21} \\ &= (0.5)(0.4) - (-0.2)(-0.2) \\ &= 0.2 - 0.04 \\ &= 0.16 \end{aligned}$$

Step 3 of 5

Calculate the $\mathbf{Z}_{11}$ parameter.

$$\begin{aligned} \mathbf{z}_{11} &= \frac{\mathbf{y}_{22}}{\Delta_y} \\ &= \frac{0.4}{0.16} \\ &= 2.5\,\Omega \end{aligned}$$

Calculate the $\mathbf{Z}_{12}$ parameter.

$$\begin{aligned} \mathbf{z}_{12} &= -\frac{\mathbf{y}_{12}}{\Delta_y} \\ &= -\frac{(-0.2)}{0.16} \\ &= \frac{0.2}{0.16} \\ &= 1.25\,\Omega \end{aligned}$$

Calculate the $\mathbf{Z}_{21}$ parameter.

$$\begin{aligned}\mathbf{z}_{21} &= -\frac{\mathbf{y}_{21}}{\Delta_y} \\ &= -\frac{(-0.2)}{0.16} \\ &= \frac{0.2}{0.16} \\ &= 1.25\ \Omega\end{aligned}$$

Calculate the $\mathbf{Z}_{12}$ parameter.

$$\begin{aligned}\mathbf{z}_{22} &= \frac{\mathbf{y}_{11}}{\Delta_y} \\ &= \frac{0.5}{0.16} \\ &= 3.125\ \Omega\end{aligned}$$

Therefore, the $\mathbf{Z}$ parameters, $\begin{bmatrix}\mathbf{z}_{11} & \mathbf{z}_{12} \\ \mathbf{z}_{21} & \mathbf{z}_{22}\end{bmatrix}$ is $\boxed{\begin{bmatrix}2.5 & 1.25 \\ 1.25 & 3.125\end{bmatrix}\Omega}$.